



# Luca Lazzaroni

Ricercatore a tempo determinato in Elettronica

DITEN — Dip. di Ingegneria Navale, Elettrica, Elettronica e delle Telecomunicazioni

SEDE Genova, Italia

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SETTORE RTD-A · IINF-01/A

AGGIORNATO In ruolo da marzo 2025

## BIBLIOMETRIA

SCOPUS

|                               |                         |                       |                        |
|-------------------------------|-------------------------|-----------------------|------------------------|
| <b>66</b><br>DOCUMENTI SCOPUS | <b>336</b><br>CITAZIONI | <b>11</b><br>INDICE H | <b>71</b><br>CO-AUTORI |
|-------------------------------|-------------------------|-----------------------|------------------------|

AGG. 17 SET 2026

### 01 Profilo

Ricercatore a tempo determinato (RTD-A) presso il Dipartimento di Ingegneria Navale, Elettrica, Elettronica e delle Telecomunicazioni (DITEN), Università degli Studi di Genova, settore IINF-01/A – Elettronica, dal marzo 2025. Dottore di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni dal 2024.

Attività di ricerca su machine learning embedded ed edge AI: toolchain TinyML, reti neurali quantizzate e binarie su microcontrollori e FPGA, percezione e decisione per la guida automatizzata, deep reinforcement learning, LLM on-device. L'attività si svolge anche nell'ambito di progetti di ricerca europei e nazionali con partner industriali.

Edge AI

TinyML

Deep learning embedded

Guida automatizzata

Deep reinforcement learning

LLM on-device

### 02 Incarichi e formazione

#### POSIZIONI

Mar 2025 – oggi **Ricercatore a tempo determinato (RTD-A)**  
DITEN, Università di Genova  
Settore scientifico-disciplinare IINF-01/A – Elettronica.

Nov 2023 – Feb 2025 **Assegnista di ricerca**  
DITEN, Università di Genova  
Assegno: “Realizzazione di modelli di Machine Learning per il trattamento di dati provenienti da veicoli autonomi”.

Nov 2020 – Ott 2023 **Dottorando di ricerca**  
DITEN, Università di Genova — XXXVI ciclo  
Borsa di dottorato: “Deep learning on the edge”.

#### FORMAZIONE

|      |                                                                                                                                                                                                                                                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | <b>Dottorato di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni</b><br>Università di Genova — conseguito il 14 marzo 2024<br>Tesi: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning” |
| 2020 | <b>Laurea Magistrale in Ingegneria Elettronica (LM-29)</b><br>Università di Genova — 24 luglio 2020<br>Tesi: “Progetto e implementazione di un motore di esecuzione multiplatforma per applicazioni IoT” · Voto: 110/110                                                                          |
| 2017 | <b>Laurea in Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)</b><br>Università di Genova — 27 ottobre 2017<br>Tesi: “Progetto model-based del controllo di robot Lego in ambiente Simulink”                                                                                           |
| 2014 | <b>Diploma di Maturità Scientifica</b><br>Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV)<br>Voto: 100/100                                                                                                                                                                           |

Abilitazione all'esercizio della professione di Ingegnere dell'Informazione — Esame di Stato superato nella seconda sessione 2020.

### 03 Progetti di ricerca

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Synergies | <b>Generation of scenarios for development, training, virtual testing and validation of CCAM systems</b><br>Europeo — HORIZON-CL5-2023-D6-01-02 · Membro dell'unità di ricerca dell'Università di Genova<br>Genera scenari di test per addestramento, test e validazione di sistemi CCAM, con focus su traffico urbano ed extraurbano — dove la conoscenza degli scenari rilevanti è molto più scarsa che in autostrada. Coinvolge le principali case automobilistiche europee e molte università del settore. |
| Hi-Drive  | <b>Addressing challenges toward the deployment of higher automation</b><br>Europeo — H2020, GA 101006664 · Membro dell'unità di ricerca dell'Università di Genova<br>Sviluppa e testa su larga scala funzionalità di guida automatizzata avanzata, passando dall'automazione condizionata (SAE L3) a quella alta, in diversi scenari e contesti stradali.                                                                                                                                                      |
| GEMINI    | <b>Green Machine Learning for the IoT</b><br>Nazionale — PRIN 2022 · Membro dell'unità di ricerca dell'Università di Genova (DII–UNITN / DITEN–UNIGE)<br>Un framework per applicazioni TinyML pienamente sostenibili distribuite lungo un continuum edge-to-cloud, pensato per dispositivi embedded senza batteria, inclusi quelli privi di connettività di rete.                                                                                                                                              |

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ADINA

### **Assistente Digitale Integrato Automotive**

Regionale — POR FESR Piemonte 2014-2020 · Membro dell'unità di ricerca dell'Università di Genova, con Lifetouch

Un assistente virtuale integrato su hardware dedicato a bordo vettura e connesso ai sistemi del veicolo: algoritmi di IA interpretano le richieste del guidatore e agiscono su funzioni veicolo e infotainment.

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## **04 Didattica e supervisione**

### **CORSI**

A.A. 2024/25, 2025/26 **Distributed Electronic Systems and Technologies for Environmental Monitoring**

Docente titolare · cod. 94631

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A.A. 2024/25, 2025/26 **Tiny Machine Learning**

Docente titolare · cod. 114752

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dall'A.A. 2022/23 **Machine Learning for Automated Driving**

Culture della materia · cod. 108716

---

A.A. 2023/24 **Computer Graphics Arch. — Games and Simulation**

Docente a contratto · cod. 172317

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A.A. 2022/23 **Autonomous Agents in Games**

Supporto alla didattica · cod. 198216

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A.A. 2020/21 **Fondamenti di Programmazione di Sistemi Elettronici**

Supporto alla didattica · cod. 166061

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A.A. 2020/21 **Approccio Makers alla Progettazione Elettronica**

Supporto alla didattica · cod. 186985

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Tutti i corsi presso l'Università degli Studi di Genova.

### **SUPERVISIONE**

- 1** Dottorando — Ali Zare, Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni (XLI ciclo)
  - 12** Tesi di Laurea — Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)
  - 10** Tesi di Laurea Magistrale — Ingegneria Elettronica (LM-29)
  - 7** Tesi di Laurea Magistrale — Ingegneria Tecnologica per la Strategia e la Sicurezza (LM/DS)
  - 2** Tesi di Laurea Magistrale — Informatica (LM-18)
- 

## **05 Curatela e servizio**

### **RUOLI EDITORIALI**

|      |                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | <b>Membro dell'Editorial Board (Early Career)</b><br>Sensors — MDPI<br>Dal 18 dicembre 2025.                                                                                  |
| 2025 | <b>Editor del volume</b><br>APPLEPIES 2025 — Springer LNEE<br>Atti della International Conference on Applications in Electronics Pervading Industry, Environment and Society. |
| 2024 | <b>Guest Editor</b><br>Energies — MDPI<br>Special Issue “Tiny Machine Learning for Energy Applications”, aprile 2024 — agosto 2025.                                           |

ORGANIZZAZIONE DI CONVEGNI

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| <b>Program Chair</b><br>APPLEPIES 2025<br>Torino, Italia, 11–12 settembre 2025.                                 |
| <b>Program Committee</b><br>APPLEPIES<br>Dal 2023.                                                              |
| <b>Program Committee</b><br>GALA — Games and Learning Alliance<br>Dal 2023.                                     |
| <b>Chairman di sessione</b><br>APPLEPIES 2023<br>1st Short Oral session — Genova, Italia, 28–29 settembre 2023. |

INCARICHI ACCADEMICI

Membro della Commissione Didattica e Ammissione del Corso di Laurea Magistrale in Engineering for Natural Risk Management.

ATTIVITÀ DI REVISIONE

| RIVISTE                                                      | CONFERENZE                                                              |
|--------------------------------------------------------------|-------------------------------------------------------------------------|
| IEEE Accessdal 2023                                          | IEEE ICRA — International Conference on Robotics and Automationdal 2026 |
| Journal of Systems Architecturedal 2023                      |                                                                         |
| Engineering Applications of Artificial Intelligence dal 2024 | IEEE/RSJ IROS — Intelligent Robots and Systems dal 2025                 |
| Sensors dal 2024                                             | GALA — Games and Learning Alliancedal 2023, 2024                        |
| IET Intelligent Transport Systems dal 2025                   | APPLEPIES dal 2021–2024                                                 |

|                                                              |          |
|--------------------------------------------------------------|----------|
| IEEE Transactions on Intelligent Vehicles                    | dal 2026 |
| IEEE Transactions on Circuits and Systems II: Express Briefs | dal 2026 |
| Robotics and Autonomous Systems                              | dal 2026 |
| Scientific Reports                                           | dal 2026 |
| Journal of Big Data                                          | dal 2026 |
| Discover Internet of Things                                  | dal 2026 |

## 06 Interventi a convegni

13 INTERVENTI

|                |                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------|
| 11-12 set 2025 | <b>APPLEPIES 2025</b><br>Torino, Italia                                                            |
| 11-14 gen 2025 | <b>43rd IEEE ICCE – International Conference on Consumer Electronics</b><br>Las Vegas, NV, USA     |
| 19-20 set 2024 | <b>APPLEPIES 2024</b><br>Torino, Italia                                                            |
| 24-26 giu 2024 | <b>55th Annual Meeting of the Italian Society of Electronics (SIE)</b><br>Genova, Italia           |
| 28-29 set 2023 | <b>APPLEPIES 2023</b><br>Genova, Italia                                                            |
| 7-9 set 2023   | <b>54th Annual Meeting of the Italian Society of Electronics (SIE)</b><br>Noto (RG), Italia        |
| 26-27 set 2022 | <b>APPLEPIES 2022</b><br>Genova, Italia                                                            |
| 7-9 set 2022   | <b>53rd Annual Meeting of the Italian Society of Electronics (SIE)</b><br>Pizzo (VV), Italia       |
| 21-22 set 2021 | <b>APPLEPIES 2021</b><br>Pisa, Italia                                                              |
| 1-2 dic 2021   | <b>GALA 2021 – Games and Learning Alliance</b><br>La Spezia, Italia (online)                       |
| 7-9 lug 2021   | <b>52nd Annual Meeting of the Italian Society of Electronics (SIE)</b><br>Trieste, Italia (online) |
| 9-10 dic 2020  | <b>GALA 2020 – Games and Learning Alliance</b><br>Laval, Francia (online)                          |

19–20 nov 2020

APPLEPIES 2020

Online

## 07 Premi e associazioni

2025

### Best Presentation of the Session — sessione “Edge Computing”

43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 11–14 gennaio 2025

A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

## ASSOCIAZIONI PROFESSIONALI

### IEEE

IEEE Member.

### IEEE Consumer Technology Society

Membro del Technical Committee “Machine Learning, Deep Learning and AI in CE (MDA)”.

### SIE — Società Italiana di Elettronica

Socio nell'Unità di Ricerca dell'Università di Genova dal 2021, area di ricerca “Sistemi elettronici ed applicazioni”.

## 08 Contatti e profili

|                                  |                                |                              |
|----------------------------------|--------------------------------|------------------------------|
| EMAIL<br>luca.lazzaroni@unige.it | UNIGE<br>rubrica.unige.it      | ORCID<br>0000-0001-8092-5473 |
| SCOPUS<br>57220892898            | GOOGLE SCHOLAR<br>w_6a6sYAAAAJ | GITHUB<br>@lucalazzaroni     |
| LINKEDIN<br>luca-lazzaroni       |                                |                              |

## 09 Pubblicazioni

72 VOCI · 66 SU SCOPUS

Tutti i valori provengono da Scopus (author ID 57220892898) e sono aggiornati automaticamente. Le 6 voci di convegni nazionali e workshop non indicizzate da Scopus sono elencate sotto, ma non concorrono a nessuno di essi.

2026

### Cross-platform deployment and characterization of an on-device generative conversational agent

P. Firpo, R. Berta, F. Bellotti, **L. Lazzaroni**, M. Fresta, A. Pighetti, A. Abedinia, N. Yoonessi, D.P. Pau

*Journal of Systems Architecture* vol. 181, art. 103981

Rivista

<https://doi.org/10.1016/j.sysarc.2026.103981>

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### A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction

L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti

*IEEE Signal Processing Letters* vol. 33, pp. 136–140

Rivista

<https://doi.org/10.1109/lsp.2025.3638676>

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### FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction

A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srour, F. Tango, F. Bellotti

*IEEE Transactions on Circuits and Systems II: Express Briefs* vol. 73, no. 8, pp. 888–892

Rivista

<https://doi.org/10.1109/tcsii.2026.3705103>

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### Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms

H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta

*IEEE Access* vol. 14, pp. 44886–44903

Rivista

<https://doi.org/10.1109/access.2026.3676567>

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### StabilEdge, Formal Verification of Edge AI Applications

**L. Lazzaroni**, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti

*IEEE Access* vol. 14, pp. 37416–37434

Rivista

<https://doi.org/10.1109/access.2026.3667991>

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### Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers

A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 321–327

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_46](https://doi.org/10.1007/978-3-032-17174-0_46)

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### Preface

M. Ruó Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero

*Lecture Notes in Electrical Engineering* pp. v–vi

Editoriale

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### Optimizing Binary AutoEncoders on STM32 Microcontrollers

H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 301–307

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_43](https://doi.org/10.1007/978-3-032-17174-0_43)

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### Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students

L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 242–248

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_34](https://doi.org/10.1007/978-3-032-17174-0_34)

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### Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform

H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 218–223

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_30](https://doi.org/10.1007/978-3-032-17174-0_30)

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### Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA

D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 25–31

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_4](https://doi.org/10.1007/978-3-032-17174-0_4)

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### Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning

O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 212–217

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_29](https://doi.org/10.1007/978-3-032-17174-0_29)

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### Implementing YoloV8 on Zynq UltraScale + MPSoC FPGA Boards

H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti

*Proceedings of SIE 2025* Lecture Notes in Electrical Engineering pp. 289–294

Nazionale

[https://doi.org/10.1007/978-3-032-17305-8\\_31](https://doi.org/10.1007/978-3-032-17305-8_31)

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### FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving

A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

*Proceedings of SIE 2025* Lecture Notes in Electrical Engineering pp. 295–301

Nazionale

[https://doi.org/10.1007/978-3-032-17305-8\\_32](https://doi.org/10.1007/978-3-032-17305-8_32)

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### Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision

A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 85–90

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_12](https://doi.org/10.1007/978-3-032-17174-0_12)

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### Embedded Deployment of an LLM-Based Voice Assistant

**L. Lazzaroni**, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 273–278

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_39](https://doi.org/10.1007/978-3-032-17174-0_39)

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### Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection

V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 286–291

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_41](https://doi.org/10.1007/978-3-032-17174-0_41)

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### A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors

S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 183–190

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_25](https://doi.org/10.1007/978-3-032-17174-0_25)

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### A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems

E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 249–254

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_35](https://doi.org/10.1007/978-3-032-17174-0_35)

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### A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents

A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 79–84

Conferenza

[https://doi.org/10.1007/978-3-032-17174-0\\_11](https://doi.org/10.1007/978-3-032-17174-0_11)

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**An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection**

L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta

*IEEE Sensors Journal* vol. 25, no. 24, pp. 44859–44874

Rivista

<https://doi.org/10.1109/jsen.2025.3627477>

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**TinyML Acceleration with MAX78000**

A. Dabbous, L. Lazzaroni, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta

*Proceedings of SIE 2024* Lecture Notes in Electrical Engineering pp. 468–474

Nazionale

[https://doi.org/10.1007/978-3-031-71518-1\\_57](https://doi.org/10.1007/978-3-031-71518-1_57)

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**Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection**

L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, L. Lazzaroni

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 156–163

Conferenza

[https://doi.org/10.1007/978-3-031-84100-2\\_19](https://doi.org/10.1007/978-3-031-84100-2_19)

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**Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking**

H. Ballout, R. Berta, I. Bochenko, L. Lazzaroni, A. Dhonthi, V. Hashemi, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 387–394

Conferenza

[https://doi.org/10.1007/978-3-031-84100-2\\_46](https://doi.org/10.1007/978-3-031-84100-2_46)

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**Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking**

F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, L. Lazzaroni

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 139–147

Conferenza

[https://doi.org/10.1007/978-3-031-84100-2\\_17](https://doi.org/10.1007/978-3-031-84100-2_17)

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**RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset**

R. Berta, F. Bellotti, M. Cossu, L. Forneris, L. Lazzaroni, A. Pighetti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 395–402

Conferenza

[https://doi.org/10.1007/978-3-031-84100-2\\_47](https://doi.org/10.1007/978-3-031-84100-2_47)

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**RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking**

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, L. Lazzaroni, C. Oh

*IEEE Sensors Letters* vol. 9, no. 9, art. 6010204

Rivista

<https://doi.org/10.1109/lensens.2025.3600982>

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**Quantization of MobileNetV2 for Resource-Constrained Microcontrollers**

L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 497–504

Conferenza

[https://doi.org/10.1007/978-3-031-84100-2\\_59](https://doi.org/10.1007/978-3-031-84100-2_59)

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**Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board**

H. Ballout, R. Berta, A. Dabbous, M. Fresta, L. Lazzaroni, A. Pighetti, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 242–249

Conferenza

[https://doi.org/10.1007/978-3-031-84100-2\\_29](https://doi.org/10.1007/978-3-031-84100-2_29)

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### On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection

M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta

*IEEE Open Journal of the Industrial Electronics Society* vol. 6, pp. 1351–1369

Rivista

<https://doi.org/10.1109/ojies.2025.3601982>

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### Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices

A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 323–330

Conferenza

[https://doi.org/10.1007/978-3-031-84100-2\\_38](https://doi.org/10.1007/978-3-031-84100-2_38)

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### Green Machine Learning for the IoT (GEMINI): A Position Paper

**L. Lazzaroni**, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli

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### Exploring Decision Transformer for Highway Automated Driving

L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh

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### Deep Learning-Based Real-Time Driver Cognitive Distraction Detection

M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta

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### Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection

M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti

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### Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices

A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti

*2025 IEEE International Conference on Consumer Electronics (ICCE)*

Digest of Technical Papers IEEE International Conference on Consumer Electronics

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<https://doi.org/10.1109/icce63647.2025.10930047>

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### Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification

A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

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### An embedded end-to-end voice assistant

**L. Lazzaroni**, F. Bellotti, R. Berta

*Engineering Applications of Artificial Intelligence* vol. 136, art. 108998

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<https://doi.org/10.1016/j.engappai.2024.108998>

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## Adversarial Policy Generation in Automated Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

*ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving*

Workshop

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## CBin-NN: An Inference Engine for Binarized Neural Networks

F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti

*Electronics* vol. 13, no. 9, art. 1624

Rivista

<https://doi.org/10.3390/electronics13091624>

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## End-to-End Dataset Collection System for Sport Activities

M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta

*Electronics* vol. 13, no. 7, art. 1286

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## YoloP-Based Pre-processing for Driving Scenario Detection

M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti

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## Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection

M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta

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## Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations

A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta

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## Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity

R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti

*Journal of Intelligent and Connected Vehicles* vol. 7, no. 3, pp. 229–244

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<https://doi.org/10.26599/jicv.2023.9210039>

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## Developing a TinyML Image Classifier in an Hour

R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti

*IEEE Open Journal of the Industrial Electronics Society* vol. 5, pp. 946–960

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## Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving

**L. Lazzaroni**, A. Pighetti, F. Bellotti, R. Berta

*Proceedings of SIE 2023* Lecture Notes in Electrical Engineering pp. 351–356

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## Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge

A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti

*IEEE Open Journal of the Industrial Electronics Society* vol. 5, pp. 781–794

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### Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach

L. Lazzaroni, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta

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### Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents

F. Ahmed, R. Berta, F. Bellotti, F. Floris, L. Lazzaroni, G. Barresi, J.R. Carrion

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2023

### Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles

L. Forneris, A. Pighetti, L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, R. Berta

*International Journal of Serious Games* vol. 10, no. 4, pp. 153–170

Rivista

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### Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild

F. Ahmed, R. Berta, F. Bellotti, L. Lazzaroni, F. Floris, G. Barresi, J.R. Carrion

*Applied Sciences* vol. 13, no. 11, art. 6432

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### Supporting a .csv-based Workflow in MongoDB for Data Analysts

M. Fresta, A. Capello, F. Bellotti, L. Lazzaroni, M. Cossu, R. Berta

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### Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, L. Lazzaroni, M. Cossu, A. De Gloria, R. Berta

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### Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, L. Lazzaroni, A. Capello, M. Cossu, A. De Gloria, R. Berta

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### Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System

M. Fresta, F. Bellotti, A. Capello, M. Cossu, L. Lazzaroni, A. De Gloria, R. Berta

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### Developing a Toolchain for Synthetic Driving Scenario Datasets

M. Cossu, R. Berta, A. Capello, A. De Gloria, L. Lazzaroni, F. Bellotti

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## Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

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## Deep Reinforcement Learning for Automated Car Parking

**L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

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2022

## Synthetic Driving Scenarios Generator

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

*53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)*

Nazionale

## Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

*53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)*

Nazionale

## Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, **L. Lazzaroni**

*53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)*

Nazionale

## An Embedded End-to-End Voice Assistant

**L. Lazzaroni**, F. Bellotti, A. De Gloria, R. Berta

*53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)*

Nazionale

## Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications

R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**

*Sensors* vol. 22, no. 3, art. 713

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## High-Level Decision-Making Non-player Vehicles

A. Pighetti, L. Forneris, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

*Games and Learning Alliance*

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## Developing a Synthetic Dataset for Driving Scenarios

J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, S. Bonora

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## Classifying Simulated Driving Scenarios from Automated Cars

M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, **L. Lazzaroni**, R. Berta

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## A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses

F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta

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2021

## Project and Development of an Italian Offline Embedded Voice Assistant

**L. Lazzaroni**, F. Bellotti, A. De Gloria, R. Berta

*52nd Annual Meeting of the Italian Society of Electronics (SIE), Trieste*

Nazionale

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## Edgine, A Runtime System for IoT Edge Applications

R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**

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## Adapting Autonomous Agents for Automotive Driving Games

G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio

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2020

## Employing an IoT Framework as a Generic Serious Games Analytics Engine

**L. Lazzaroni**, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

*Games and Learning Alliance*

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

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